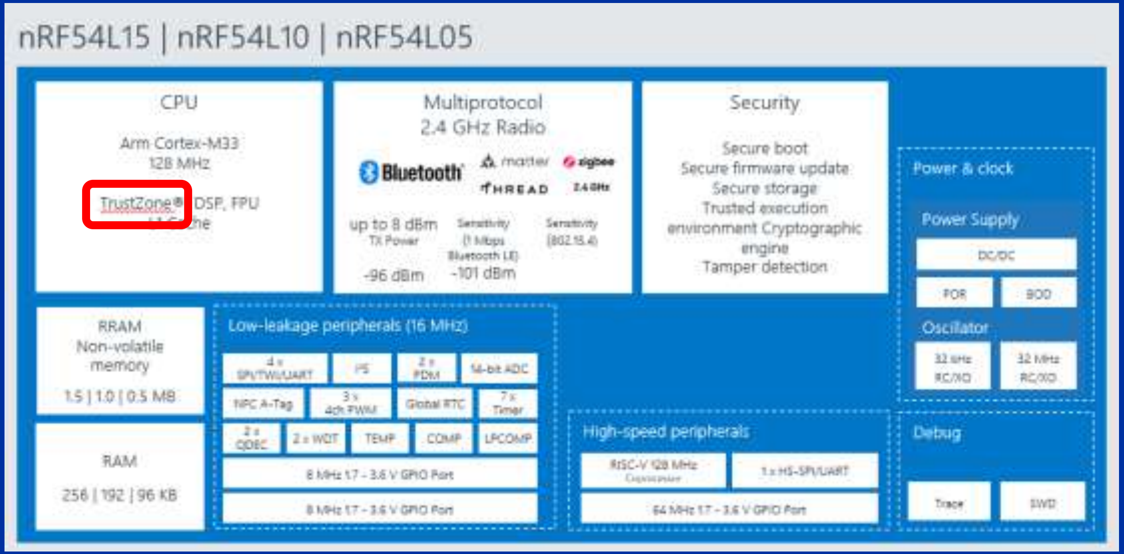
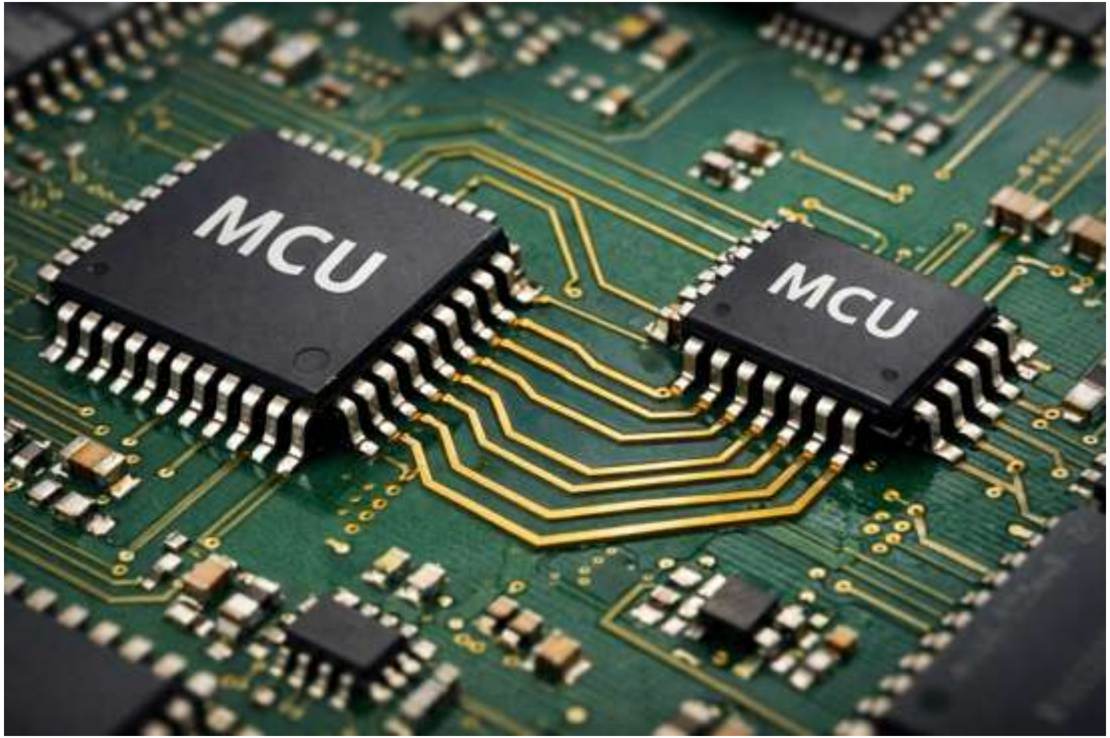
nRF54L Series Trustzone



nRF54L Deep-Dive

TrustZone

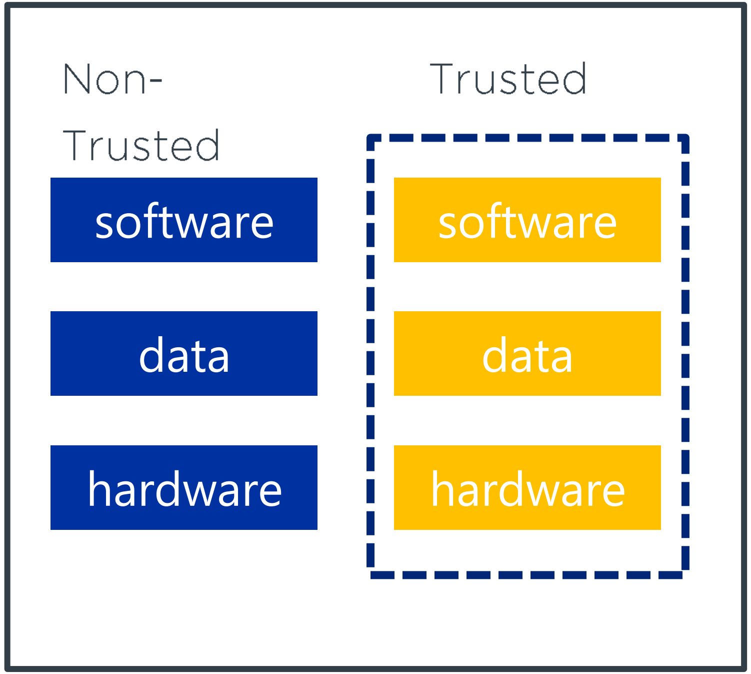
How do I protect specific software modules or data?



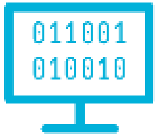
TrustZone

- <u>Two-World Principle:</u>
  The processor is divided into a “Secure World” (secure) and a “Non-Secure World” (normal).

- <u>Isolation:</u>
  Sensitive data such as passwords, cryptographic keys, or … is processed in the secure world. Even if the operating system is hacked in the normal world, access to the secure world remains blocked.



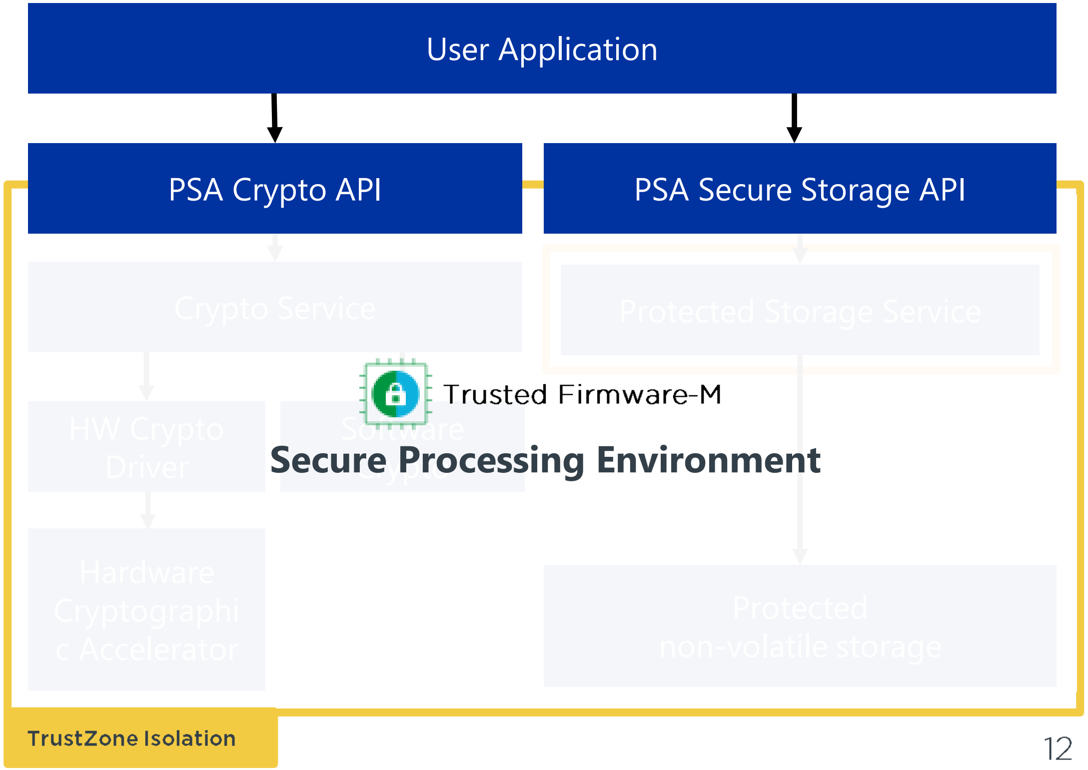
Implement – Firmware



PSA APIs

Standardised interface towards security services.

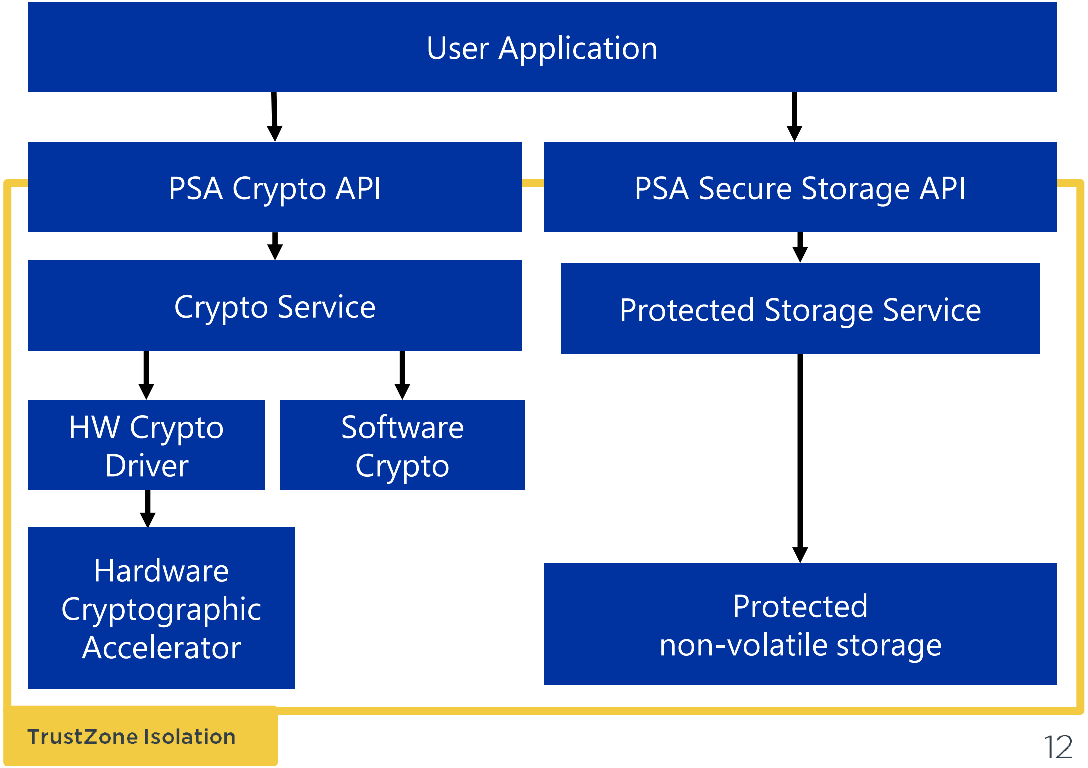
- **<u>Enhanced security</u> - enabling devices to meet industry-standard security**
- **<u>Implementation agnostic</u> - Abstracts hardware/software implementation differences between platforms.**
- **<u>Flexible and scalable</u> – Supports a variety of use cases, from simple to complex systems**
- **<u>Future proof</u> – PSA APIs are designed to be updated as security threats evolve**



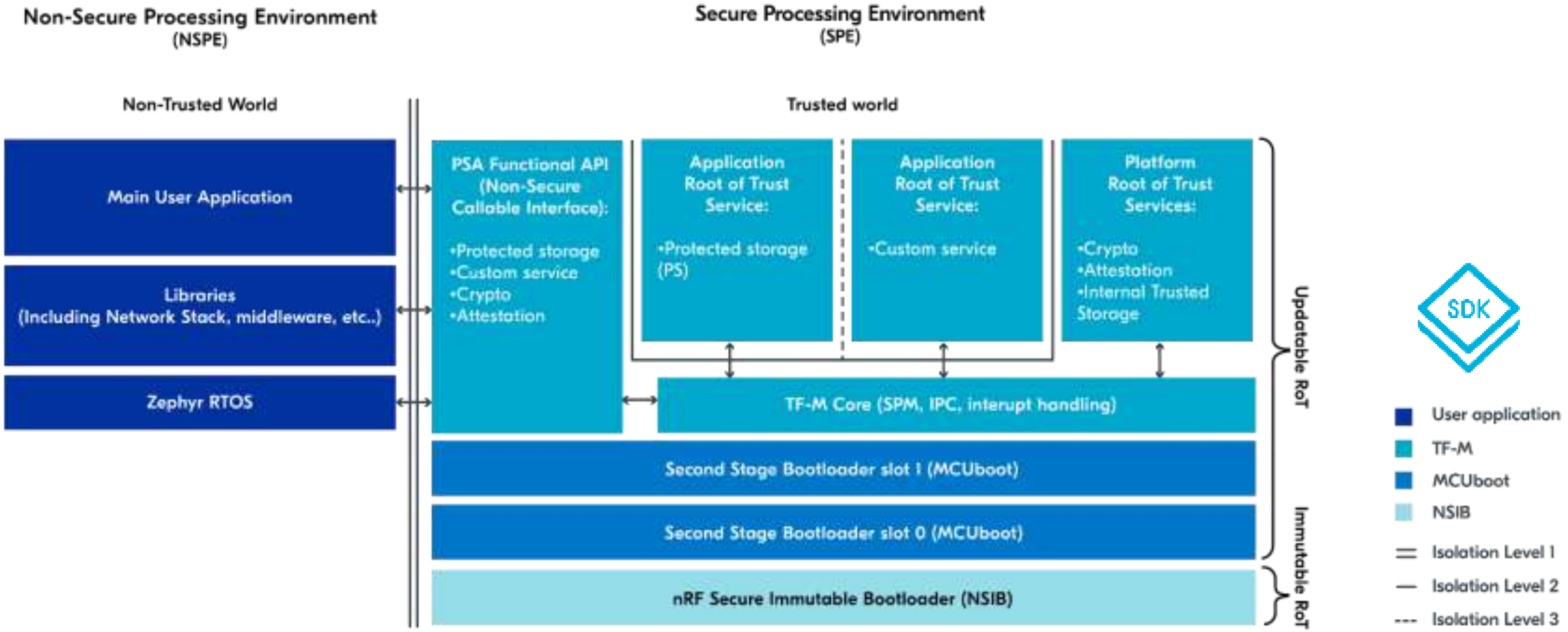
Implement – Firmware

Trusted Firmware-M (TF-M)

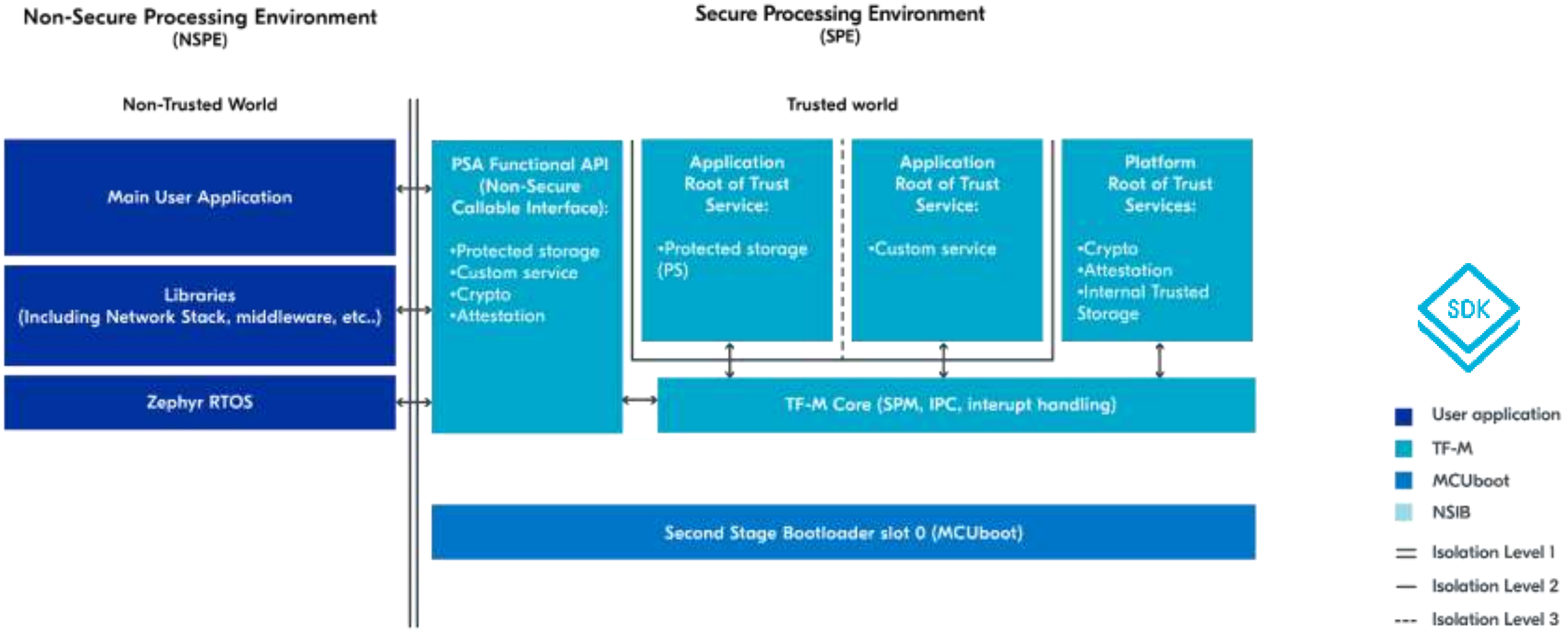
- **TF-M is a secure processing environment**
- **Runs on TrustZone enabled hardware**
- **Isolates critical security services and data from non-secure user application**
- **Provides PSA Root of Trust and secure services implementation, such as:**
    - **Cryptographic services**
    - **Secure storage**
    - **Attestation service**



TF-M in nRF Connect SDK

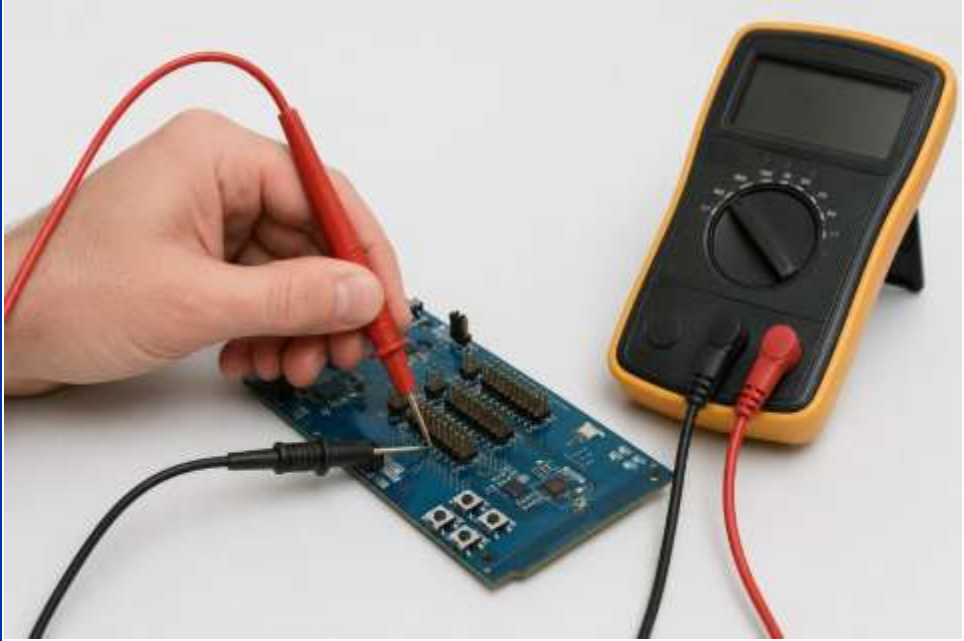


TF-M in nRF Connect SDK



Hands-on

- Using Trustzone
  - Getting familiar with Trustzone Secure/Non-Secure Separation
  - Using Secure PSA API
  - PSA Secure Storage on TrustZone



Step 1 - Explore the Project Structure

1. Download the lab project from here
2. Open the project.

   💡 **Note:** We will *Add build configuration* later on!
3. Review the project directory structure.
4. Identify the following files:
   - prj.conf
   - src/main.c
   - boards/nrf54l15dk_nrf54l15_cpuapp_ns.overlay
5. Discuss the purpose of each file.

Step 2 - Select the Correct Board Target

6. Compare the following board targets:
    - nrf54l15dk/nrf54l15/cpuapp
    - nrf54l15dk/nrf54l15/cpuapp/ns

7. Discuss the differences.

8. Now, **Add build configuration**. Use following settings:

    Board Target: nrf54l15dk/nrf54l15/cpuapp/ns

Step 3 - Build the TrustZone Application

9. Open the terminal to check the outputs during build process.

10. Observe the build output.

11. Identify the Secure and Non-secure images produced during the build.

Step 4 - Review the TrustZone Configuration

12. Open prj.conf.
13. Locate the TF-M related configuration options.
14. Identify the selected TF-M profile.
15. Discuss why the Secure and Non-secure worlds cannot share the same peripheral simultaneously.

Step 5 - Flash and Observe TrustZone

16. Flash the firmware
17. Open two Serial Terminals.
18. Connect one terminal to the Secure UART output. (VCOM 1)
19. Connect the second terminal to the Non-secure application console. (VCOM 0)
20. Reset the board.
21. Observe the Secure boot messages.
22. Observe the Non-secure application output.

Step 6 - Execute a Secure PSA Crypto Service

23. Inspect the PSA Crypto function call in main.c.

24. Build and flash the application if necessary.

25. Observe the application output.

26. Verify that random data is successfully returned.

27. Discuss where the cryptographic operation actually executes.

Step 7 - PSA Secure Storage on TrustZone

28. Download the lab project from here
29. Open the project within VS code
30. What happens if you forget to set `CONFIG_TFM_PARTITION_STORAGE=y` ?
31. Which PSA Secure Storage calls are used?